

Enterprise Google Cloud Data Platform Blueprint

Volume 07: Security Blueprint

Zero Trust, control mapping and regulated-workload security

Document attribute	Value
Version	0.1 reference implementation
Date	2026-07-10
Intended audience	Executive, architecture, security, platform engineering, data, AI, SRE and FinOps stakeholders
Scope	Fortune 100 reference blueprint for petabyte-scale Google Cloud data platform
Status	Generated implementation starter set; verify service-specific settings against current Google Cloud documentation

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Inherited Fortune 100 Assumptions

Area	Assumption
Scale	10-50 PB historical data; 50-250 TB/day new data; billions of events/day
Organization	Global enterprise with central platform team and federated domain teams
Regulatory	GDPR, HIPAA, PCI DSS, SOC 2, ISO 27001, NIST and regional residency constraints may apply
Availability	Tier 0 RTO <1 hour and RPO <5 minutes; Tier 1 RTO <4 hours and RPO <15 minutes
Cloud posture	Hybrid and multi-cloud integration with Google Cloud as the enterprise data platform foundation
Security	Zero Trust, least privilege, private connectivity, service perimeters, CMEK for restricted data

1. Security Strategy

The blueprint applies Zero Trust through identity, device, network, workload, data and service-control layers. Every workload is authenticated, authorized, encrypted, logged and monitored.

2. Control Architecture

Controls include IAM, Organization Policies, VPC SC, Access Context Manager, KMS, Secret Manager, Sensitive Data Protection, SCC and centralized audit logging.

Service	Purpose	Benefits	Limitations / Pitfalls	Cost Controls
BigQuery	Enterprise analytics warehouse and curated serving layer	Serverless SQL, separation of storage/compute, strong ecosystem	Poor partitioning and unrestricted ad-hoc queries can create high costs	Reservations, labels, max bytes billed, partitioning, clustering
Cloud Storage	Raw, landing, archive and open-format lakehouse storage	Durable object storage, lifecycle policies, broad integration	Bucket sprawl and direct consumer access can weaken governance	Lifecycle policies, storage classes, retention controls
BigLake	Governed access to Cloud Storage data through BigQuery interfaces	Fine-grained access without direct bucket reads	Requires metadata and permission design	Table lifecycle, open-format governance
Dataflow	Batch and streaming pipelines using Apache Beam	Unified model, autoscaling, managed execution	Complex stateful pipelines require strong testing	Autoscaling, worker sizing, streaming engine
Pub/Sub	Event ingestion and decoupling	Reliable async messaging, retry and dead-letter support	Schema/version drift can impact consumers	Retention tuning and payload design
Knowledge Catalog	Metadata, glossary, lineage, governance and data discovery	Enterprise context layer for data and AI	Incomplete stewardship reduces trust	Metadata ownership and lifecycle controls

Figure: Security and Zero Trust Control Plane

Security and Zero Trust Control Plane

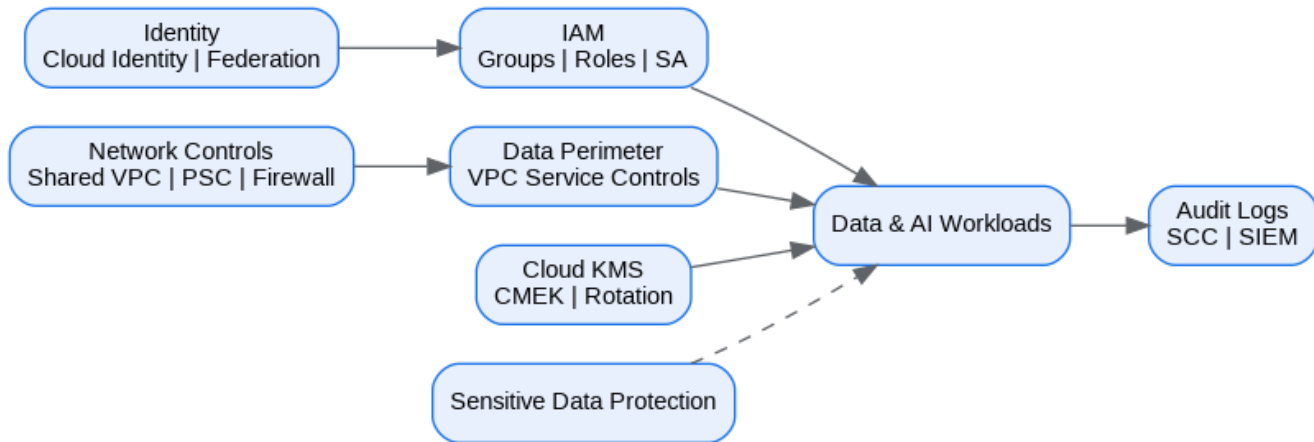
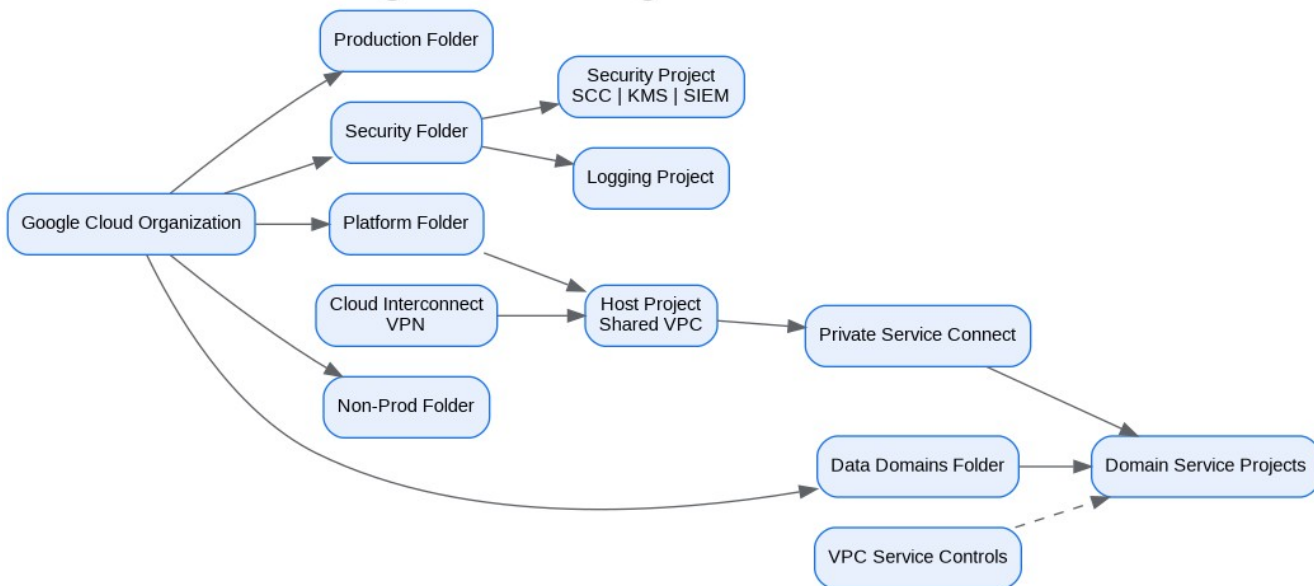


Figure: Google Cloud Landing Zone and Shared VPC

Google Cloud Landing Zone and Shared VPC



3. Regulatory Mapping

NIST, HIPAA, GDPR, PCI DSS, SOC 2 and ISO 27001 controls are mapped to platform capabilities and validation evidence.

Regulatory Theme	Platform Control	Evidence
Access control	IAM groups, custom roles, JIT access	IAM policy export and audit logs
Encryption	KMS/CMEK, TLS, private access	KMS inventory and key-rotation logs
Auditability	Cloud Audit Logs and SIEM integration	Log retention and query evidence
Data privacy	Classification, DLP, masking	Scan reports and policy tags
Exfiltration prevention	VPC SC, egress controls	Perimeter policy and violation logs

4. Threat Model

Primary threats include credential misuse, privilege escalation, accidental data exposure, cross-region residency breach, model/prompt leakage and supply-chain compromise.

5. Security Readiness

All production services require threat modeling, control mapping, penetration-test scope, logging validation and incident-response runbook.

Architecture Decision Records

ADR	Decision	Problem	Alternatives	Decision	Tradeoffs/Risks	Mitigation
ADR-001	Google Cloud as enterprise data platform	Fragmented analytics estate	Continue legacy; multi-cloud split; Google Cloud data platform	Adopt Google Cloud data platform with hybrid integration	Requires operating-model change	Phased landing zone, domain pilots, and governance automation
ADR-002	BigQuery analytical core	Need PB-scale SQL analytics	Legacy EDW; Spark-only; third-party warehouse	Use BigQuery for curated/serving layers	Cost governance required	Reservations, labels, query controls, FinOps dashboards
ADR-003	Lakehouse with Cloud Storage and BigLake	Need open data and governed access	BigQuery-only; unmanaged object lake	Use Cloud Storage + BigLake + BigQuery	Open table governance maturity required	Catalog, policies, certified data-product lifecycle
ADR-004	Zero Trust and VPC SC	Exfiltration risk and regulated data	Network-only perimeter; IAM-only controls	Use identity, network, data and service perimeters	Service exceptions require testing	Perimeter dry runs and explicit ingress/egress policies

Risks and Mitigations

Risk	Impact	Mitigation
Underestimated legacy complexity	High	Complete source, pipeline, report and dependency assessment before migration waves
Data ownership gaps	High	Create domain owner RACI and data-product certification gate
Cost overrun	High	Enforce labels, budgets, reservations, query guardrails and monthly FinOps reviews
Security control drift	Critical	Deploy IaC, policy-as-code, automated IAM review and security posture monitoring
AI data leakage	Critical	Use certified datasets, service perimeters, DLP, RAG guardrails and model governance

Production Readiness Checklist

- All projects, networks, KMS keys and service accounts deployed through Terraform.
- All data products registered with owner, steward, classification, SLO, contract and support model.
- All restricted paths validated against VPC Service Controls and private connectivity requirements.
- All pipelines implement retries, dead-letter handling, reconciliation and operational metrics.
- All dashboards, alerts, runbooks and escalation routes tested before production cutover.
- All cost allocation labels present and billing export dashboards available to product owners.

References

- Google Cloud Architecture Framework: <https://docs.cloud.google.com/architecture/framework>
- Google Cloud Enterprise Foundations Blueprint: <https://docs.cloud.google.com/architecture/blueprints/security-foundations>
- Google Cloud Terraform blueprints and modules:
<https://docs.cloud.google.com/docs/terraform/blueprints/terraform-blueprints>
- Build data products in a data mesh: <https://docs.cloud.google.com/architecture/build-data-products-data-mesh>
- Architecture and functions in a data mesh: <https://docs.cloud.google.com/architecture/data-mesh>
- Knowledge Catalog formerly Dataplex: <https://cloud.google.com/products/knowledge-catalog>
- Optimize costs with FinOps hub: <https://docs.cloud.google.com/billing/docs/how-to/finops-hub>
- VPC Service Controls supported products: <https://docs.cloud.google.com/vpc-service-controls/docs/supported-products>